Net-Zero Strategies for Mexico City: Addressing Climate Action Plan Gaps

Vilas Annavarapu

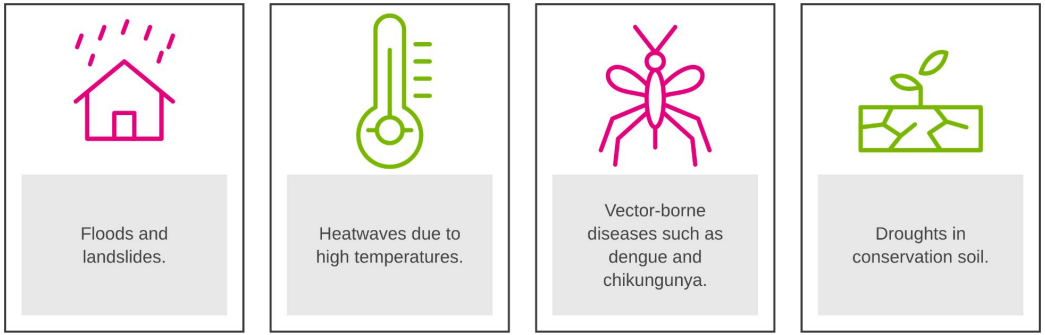
Strategy to Achieve Net-Zero Climate Goals



Gaps in Mexico City's Climate Action Plan

- Invest in bicycles, e-bikes, and walkability
  - Increase costs for ICEs
  - Support EV adoption, but prioritize public transport
- Leverage the potentials of geothermal energy
- Promote heat pumps
  - Support building retrofits
  - Mandate installation in new buildings

Main Adaptation Challenges Facing Mexico City



CNN
https://www.cnn.com · 3 Jun 2025

Record rain floods Mexico City, traps people

Mexico City was hit with record rainfall that didn't relent for more than five

Extreme heat killing more than 100 people in Mexico hotter and much more likely due to climate change

CityLab | Environment

How Mexico City Averted All-Out Drought

With just-in-time rain and a looming presidential election, Mexico City never reached 'Day Zero.' But the politicized threat helped propel water infrastructure as a priority policy issue.

By Maya Averbuch

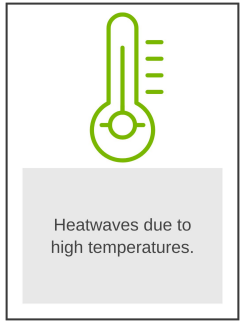
October 14, 2024 at 8:30 PM GMT+8

Strategies to Address Adaptation Challenges



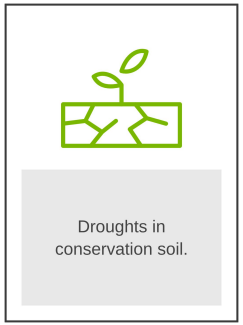
Floods and landslides.

- Permeable concrete
- Improve drainage infrastructure
- Develop controlled flood ozones



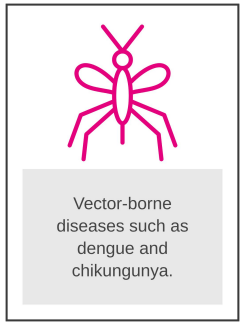
Heatwaves due to high temperatures.

- Increase tree cover
- Building design oriented for cooling
- Shift to late night and early morning activities



Droughts in conservation soil.

- Citywide rainwater harvesting
- Household water limits
- Invest in desalination technologies



Vector-borne diseases such as dengue and chikungunya.

- Disease education
- Medicine and vaccine distribution infrastructure
- PPE stockpiles

Financial Implications

- More bicycles → less road improvement costs
- More public transit → increased public maintenance costs
- Geothermal energy → investment $ can't go elsewhere (ex. healthcare)
- Heat Pumps → additional costs for developers



Infrastructure, tech, and public good investments: divert city $ from other projects

Equity Considerations

- Access to non-car forms of transportation → increases equity
- Public good investments → increases equity
- Additional building restrictions → increased housing prices & decreased equity; financial burdens during household electrification transition
- Investments in clean energy (ex. geothermal) → long-term cost decreases

Risks

- Political resistance to car restrictions
- Infrastructure, investment, and public infrastructure are expensive
- Unsuccessful technology investments → wasted $

Solutions

- Campaigns about health benefits of walking and biking
- Cheaper now than 10 years from now
- Potential to develop profitable industry with jobs

Thank You